

RAG Configuration Performance Report

Evaluation Framework: DeepEval Stack

Date: October 1, 2026

Target Model: GPT-4o

Executive Summary

This tuning matrix evaluates three distinct Retrieval-Augmented Generation (RAG) parameter tiers against a benchmark Golden dataset. Metrics capture the core performance trade-offs of the RAG Triad: generation quality (Faithfulness, Answer Relevancy) and retrieval efficiency (Contextual Recall, Contextual Relevancy).

Performance Evaluation Matrix

Config Tier	Answer Relevancy	Faithfulness	Contextual Recall	Contextual Relevancy	Composite Score
Simple (MVP)	0.88	0.96	0.62	0.94	0.854
Medium (Standard)	0.92	0.94	0.89	0.85	0.912
Heavy (Deep)	0.85	0.89	0.95	0.64	0.855

Tested Parameter Layouts

- **Simple (MVP):** k=3, temperature=0.0, top_p=0.1, max_tokens=256
- **Medium (Standard Enterprise):** k=6, temperature=0.2, top_p=0.8, max_tokens=1024
- **Heavy (Deep Analytics):** k=15, temperature=0.4, top_p=0.95, max_tokens=4096

Key Optimization Insights

1. **The Medium Sweet Spot:** The Medium configuration achieved the highest mathematical Composite Score (0.912). It provides the optimal trade-off by securing high Contextual Recall without introducing significant noisy context block clutter.
2. **Heavy Tier Context Bloat:** Scaling retrieval depth to k=15 successfully maximized Contextual Recall (0.95), but severely degraded Contextual Relevancy (0.64) and dropped Faithfulness. This indicates that excessive noise introduced into the prompt wrapper risks distracting the generator and inducing hallucinations.